

2022 AIME I Problems/Problem 13

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Problem

Let S be the set of all rational numbers that can be expressed as a repeating decimal in the form $0.\overline{abcd}$, where at least one of the digits a , b , c , or d is nonzero. Let N be the number of distinct numerators obtained when numbers in S are written as fractions in lowest terms. For example, both $\frac{4}{11}$ and $\frac{410}{3333}$ are counted among the distinct numerators for numbers in S because $0.\overline{3636} = \frac{4}{11}$ and $0.\overline{1230} = \frac{410}{3333}$. Find the remainder when N is divided by 1000.

Solution 1

$$0.\overline{abcd} = \frac{abcd}{9999} = \frac{x}{y}, 9999 = 9 \times 11 \times 101.$$

Then we need to find the number of positive integers x that (with one of more y such that $y|9999$) can meet the requirement $1 \leq x \cdot \frac{9999}{y} \leq 9999$.

Make cases by factors of x . (A venn diagram of cases would be nice here.)

Case A :

$3 \nmid x$ and $11 \nmid x$ and $101 \nmid x$, aka $\gcd(9999, x) = 1$.

Euler's totient function counts these:

$$\varphi(3^2 \cdot 11 \cdot 101) = ((3 - 1) \cdot 3)(11 - 1)(101 - 1) = \mathbf{6000}$$

values (but it's enough to note that it's a multiple of 1000 and thus does not contribute to the final answer)

Note: You don't need to know this formula. The remaining cases essentially re-derive the same computation for other factors of 9999. This case isn't actually different.

The remaining cases have 3 (or 9), 11, and/or 101 as factors of $abcd$, which cancel out part of 9999. Note: Take care about when to use 3 vs 9.

Case B : $3|x$, but $11 \nmid x$ and $101 \nmid x$.

Then $abcd = 9x$ to leave 3 uncanceled, and $x = 3p$, so $x \leq \frac{9999}{9} = 1111$, giving:

$$x \in 3 \cdot \left\{1, \dots, \left\lfloor \frac{1111}{3} \right\rfloor\right\},$$

$$x \notin (3 \cdot 11) \cdot \left\{1 \dots \left\lfloor \frac{1111}{3 \cdot 11} \right\rfloor\right\},$$

$$x \notin (3 \cdot 101) \cdot \left\{1 \dots \left\lfloor \frac{1111}{3 \cdot 101} \right\rfloor\right\},$$

$$\text{for a subtotal of } \left\lfloor \frac{1111}{3} \right\rfloor - \left(\left\lfloor \frac{1111}{3 \cdot 11} \right\rfloor + \left\lfloor \frac{1111}{3 \cdot 101} \right\rfloor \right) = 370 - (33 + 3) = \mathbf{334} \text{ values.}$$

Case *C*: $11|x$, but $3 \nmid x$ and $101 \nmid x$.

$$\text{Much like previous case, } abcd \text{ is } 11x, \text{ so } x \leq \frac{9999}{11} = 909,$$

$$\text{giving } \left\lfloor \frac{909}{11} \right\rfloor - \left(\left\lfloor \frac{909}{11 \cdot 3} \right\rfloor + \left\lfloor \frac{909}{11 \cdot 101} \right\rfloor \right) = 82 - (27 + 0) = \mathbf{55} \text{ values.}$$

Case *D*: $3|x$ and $11|x$ (so $33|x$), but $101 \nmid x$.

$$\text{Here, } abcd \text{ is } 99x, \text{ so } x \leq \frac{9999}{99} = 101,$$

$$\text{giving } \left\lfloor \frac{101}{33} \right\rfloor - \left\lfloor \frac{101}{33 \cdot 101} \right\rfloor = 3 - 0 = \mathbf{3} \text{ values.}$$

Case *E*: $101|x$.

$$\text{Here, } abcd \text{ is } 101x, \text{ so } x \leq \frac{9999}{101} = 99,$$

$$\text{giving } \left\lfloor \frac{99}{101} \right\rfloor = \mathbf{0} \text{ values, so we don't need to account for multiples of } 3 \text{ and } 11.$$

To sum up, the answer is

$$6000 + 334 + 55 + 3 + 0 \equiv \boxed{392} \pmod{1000}.$$

Clarification

In this context, when the solution says, "Then $abcd = 9x$ to leave 3 uncanceled, and $x = 3p$," it is a bit vague. The best way to clarify this is by this exact example - what is really meant is we need to divide by 9 first to achieve 1111, which has no multiple of 3; thus, given that the fraction x/y is the simplest form, x can be a multiple of 3.

Similar explanations can be said when the solution divides 9999 by 11, 101, and uses that divided result in the PIE calculation rather than 9999.

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Sketch of Constructive Counting Solution

If anyone wants to write this up, basically I thought to do:

$$\sum_{d|9999} \varphi(d) = 9999$$

and which is 9999 (identity), but then we must subtract all the numbers that show up multiple times in the numerator. We do this by doing casework on n , like if $1111 < n \leq 3333$ we check how many n are coprime to 3333 in this range, giving the amount of n that show up multiple times in both $\frac{n}{3333}$ and $\frac{n}{9999}$. repeat this bounding process and it should work.

Video Solution

<https://youtu.be/0FZyjuIOHnA>

<https://MathProblemSolvingSkills.com>

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